

Terratrace Project Unit Test Report

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Project scope: whole Terratrace project unit-test surface across frontend and backend workspaces.

Unit-test definition: isolated utilities, components, hooks, and middleware tested with mocked network, router, database, or framework boundaries. Backend route/database integration tests remain documented in test/integration-test.docx.

1. Execution Summary

Suite	Command Executed	Environment	Test Files	Tests	Status	Notes
Frontend project unit tests	npm run test -- workspace=frontend	Vitest v4.1.5, jsdom, React Testing Library	11 passed / 11	52 passed / 52	Pass	Runner emitted localStorage and jsdom scrollTo warnings; they did not fail assertions.
Backend project unit tests	npm run test -- workspace=backend -- src/utls/auth.test.ts src/utls/analytics.test.ts src/middleware/auth.test.ts	Vitest v4.1.5, Node.js	2 passed, 1 failed / 3	14 passed, 1 failed / 15	Fail	analytics.test.ts failed because carbonSavedKg expected 42 but actual value was 41.9.

Overall result: 66 of 67 unit tests passed. One backend analytics utility assertion failed in the current workspace.

2. Module Results

Module Section: General

Objective: Validate the general authentication features from the feature matrix: register new user account, login, logout/session helper behavior, and session validation boundaries.

Command: npm run test --workspace=frontend -- AuthForm LoginPage SignupPage; npm run test -- workspace=backend -- src/utls/auth.test.ts src/middleware/auth.test.ts

Result: Frontend auth/page files passed; backend auth helper and middleware files passed. Documented rows summarize the isolated unit assertions for general auth behavior.

Isolation and mocking: Next.js navigation, auth utilities, bcrypt/JWT helpers, and Express request/response objects are mocked or controlled. Components are rendered with Testing Library and backend helpers are invoked directly.

Test Case ID	Unit Under Test	Scenario	Status	Evidence
UT-AUTH-001	AuthForm	Login mode renders email, password, and submit controls.	Pass	DOM assertions found expected controls.
UT-AUTH-002	AuthForm	Empty login submission is blocked.	Pass	onSubmit was not called.
UT-AUTH-003	AuthForm	Short login password is rejected.	Pass	Validation message rendered and callback was not called.
UT-AUTH-004	AuthForm	Mismatched signup passwords are rejected.	Pass	Passwords do not match message rendered.
UT-AUTH-005	AuthForm	Valid login submits normalized email and password.	Pass	Callback received expected arguments.

Test Case ID	Unit Under Test	Scenario	Status	Evidence
UT-AUTH-006	AuthForm	Signup-only username, confirm password, and role fields render.	Pass	Signup fields were present.
UT-AUTH-007	LoginPage	Successful login redirects to role-aware dashboard path.	Pass	Router push and refresh were called.
UT-AUTH-008	LoginPage	Failed login displays API error and avoids navigation.	Pass	Error text rendered; router push was not called.
UT-AUTH-009	LoginPage	Static login page heading, inputs, button, and signup link render.	Pass	All static elements were found.
UT-AUTH-010	SignupPage	Successful signup calls auth helper and redirects.	Pass	signUp, router push, and refresh matched expected calls.
UT-AUTH-011	SignupPage	Signup API errors stay on the page.	Pass	Error message rendered and router push was not called.
UT-BACKEND-AUTH-001	auth helpers	Hash password and verify only matching password.	Pass	bcrypt hash was not raw password; correct password passed and wrong password failed.
UT-BACKEND-AUTH-002	auth helpers	Reject empty or non-bcrypt stored hashes.	Pass	Invalid hashes resolved false.
UT-BACKEND-AUTH-003	auth helpers	Sign and verify a session token.	Pass	Verified session user matched expected payload.
UT-BACKEND-AUTH-004	auth helpers	Reject malformed and expired session tokens.	Pass	Both invalid tokens threw errors.
UT-BACKEND-AUTH-005	requireAuth	Reject request without session token.	Pass	Response returned 401 missing session and next was not called.
UT-BACKEND-AUTH-006	requireAuth	Attach user and continue for valid bearer session.	Pass	req.user was populated and next was called once.
UT-BACKEND-AUTH-007	requireAuth	Reject invalid or expired session token.	Pass	Response returned 401 invalid or expired session.
UT-GENERAL-GAP-001	Logout UI	Logout sidebar actions are implemented, but no isolated unit row is currently documented for the logout click path.	Not covered	Feature matrix row 3 is covered by implementation evidence, but the current unit suite has no logout-specific test case.

Module Section: Profile Management

Objective: Validate profile management features from the feature matrix: view profile, update profile, delete/deactivate account, and change password.

Command: `npm run test --workspace=frontend -- useUser ProfileTab`

Result: 2 frontend files passed, 11 documented profile tests passed, 0 failed.

Isolation and mocking: React Query hooks, user mutations, and profile component state are mocked or controlled. Components are rendered with Testing Library and hook behavior is verified through a local query client.

Test Case ID	Unit Under Test	Scenario	Status	Evidence
UT-PROFILE-001	useUser	Current-user hook returns mocked authenticated user.	Pass	Hook data matched mocked user.
UT-PROFILE-002	useUpdateUser	Profile update invalidates cached user query.	Pass	invalidateQueries was called for user query.
UT-PROFILE-003	useDeleteAccount	Account deletion clears query cache.	Pass	Query client clear was called.
UT-PROFILE-004	useChangePassword	Password-change API errors propagate.	Pass	Mutation rejected with expected message.
UT-PROFILE-005	ProfileTab	Profile identity fields render from user data.	Pass	Username, email, and role label were visible.
UT-PROFILE-006	ProfileTab	Blank username is rejected.	Pass	Validation message rendered and update mutation was not called.
UT-PROFILE-007	ProfileTab	Unchanged username skips unnecessary update.	Pass	Update mutation was not called.
UT-PROFILE-008	ProfileTab	Changed username calls mutation and shows success feedback.	Pass	Mutation payload and success text matched.
UT-PROFILE-009	ProfileTab	Password form requires all fields.	Pass	Validation message rendered.
UT-PROFILE-010	ProfileTab	Password API error is displayed.	Pass	API error message rendered.
UT-PROFILE-011	ProfileTab	Delete confirmation button stays disabled until DELETE is entered.	Pass	Button disabled/enabled state matched input.

Module Section: Eco-friendly Directory Module

Objective: Validate eco-directory features from the feature matrix: sustainable directory display, geo-specific search, and favourites.

Command: `npm run test --workspace=frontend -- locationFilters SearchBar PlaceCard EcoDirectoryClient useFavourites`

Result: 5 frontend files passed, 20 documented directory and favourites tests passed, 0 failed.

Isolation and mocking: Location hooks, router functions, image subcomponents, catalog data, auth state, fetch, localStorage, mutation responses, and QueryClient invalidation are controlled. Pure filter utilities are tested directly.

Test Case ID	Unit Under Test	Scenario	Status	Evidence
UT-ECO-001	locationFilters	Normalize empty, all, and whitespace-only filters.	Pass	Returned empty normalized filter values.
UT-ECO-002	locationFilters	Trim valid filter values before building API URL.	Pass	URL contained encoded query, city, and category parameters.
UT-ECO-003	locationFilters	Omit blank query parameters.	Pass	URL had no query string.
UT-ECO-004	locationFilters	Read only supported categories from search params.	Pass	Unsupported category fell back to all.
UT-ECO-005	SearchBar	Submit trimmed search query.	Pass	onSubmit received trimmed text.
UT-ECO-006	SearchBar	Debounce trimmed query changes.	Pass	onChange received trimmed text after debounce.
UT-ECO-007	SearchBar	Clear input and notify handlers.	Pass	Input cleared and callbacks received empty string.
UT-ECO-008	PlaceCard	Render place details and eco certifications.	Pass	Name, city, category, and certifications rendered.
UT-ECO-009	PlaceCard	Link to encoded eco-directory detail route.	Pass	href matched encoded route.
UT-ECO-010	EcoDirectoryClient	Render category counts and first page from mocked data.	Pass	Counts, cards, and page indicator matched.
UT-ECO-011	EcoDirectoryClient	Move to next page when pagination is clicked.	Pass	Next page item and page indicator rendered.
UT-ECO-012	EcoDirectoryClient	Update URL when search, city, and category filters change.	Pass	Router replace calls matched expected URLs.
UT-ECO-013	EcoDirectoryClient	Show loading and error states without catalog.	Pass	Loading and error states rendered.
UT-ECO-014	EcoDirectoryClient	Show empty state and clear active filters.	Pass	Empty text rendered and filters routed back to /eco-directory.
UT-FAV-001	useFavourites	Logged-out user receives empty array when localStorage is empty.	Pass	Hook returned empty array.
UT-	useFavourites	Logged-out user receives	Pass	Hook returned stored local

Test Case ID	Unit Under Test	Scenario	Status	Evidence
FAV-002		localStorage favourites.		favourite items.
UT-FAV-003	useFavourites	Hook data updates when a local favourite is added.	Pass	Rendered hook data reflected local change.
UT-FAV-004	useSyncFavourites	Sync does nothing when user is logged out.	Pass	No upload request was attempted.
UT-FAV-005	useSyncFavourites	Successful sync uploads all favourites and clears local storage.	Pass	Upload calls succeeded and local storage was cleared.
UT-FAV-006	useSyncFavourites	Partial sync failure retains only failed favourites.	Pass	Failed favourite stayed in local storage.

Module Section: Green Itinerary Module

Objective: Track unit-test coverage for itinerary creator, smart recommendations, and weather integration features from the feature matrix.

Command: No dedicated isolated unit command is currently documented for this module.

Result: No unit-test rows are currently documented for this module in test/test-list.md or test/unit-test.docx.

Isolation and mocking: Not applicable until module-specific unit tests are added. Backend route/API behavior for these areas should remain in integration documentation when it crosses HTTP or database boundaries.

Test Case ID	Unit Under Test	Scenario	Status	Evidence
UT-ITINERARY-GAP-001	TripsTab / useTrips / trips utilities	Itinerary creator has implementation evidence in the feature matrix, but no isolated unit test row is currently documented.	Not covered	No TripsTab, useTrips, or itinerary utility test appears in test/test-list.md.
UT-ITINERARY-GAP-002	Smart recommendation UI/hooks/utilities	Smart recommendations have implementation evidence in the feature matrix, but no isolated unit test row is currently documented.	Not covered	No smart recommendation unit test appears in test/test-list.md.
UT-ITINERARY-GAP-003	Weather forecast UI/hooks/utilities	Weather integration has implementation evidence in the feature matrix, but no isolated unit test row is currently documented.	Not covered	No weather unit test appears in test/test-list.md.

Module Section: Carbon Footprint Module

Objective: Track unit-test coverage for emission calculator, impact insights, and offset integration features from the feature matrix.

Command: No dedicated isolated unit command is currently documented for this module.

Result: No carbon footprint unit-test rows are currently documented in test/test-list.md or test/unit-test.docx.

Isolation and mocking: Not applicable until carbon calculator/component tests are added. Analytics utility tests are listed under the Analytics and Reporting Module because they validate dashboard impact summaries.

Test Case ID	Unit Under Test	Scenario	Status	Evidence
UT-CARBON-GAP-001	Carbon calculator utilities/components	Emission calculator has implementation evidence in the feature matrix, but no isolated carbon calculator unit test row is currently documented.	Not covered	No frontend carbonFootprint or backend carbonCalculator test appears in test/test-list.md.
UT-CARBON-GAP-002	Impact insight components	Impact insights have implementation evidence in the feature matrix, but no isolated component test row is currently documented.	Not covered	No ImpactInsights or carbon result component unit test appears in test/test-list.md.
UT-CARBON-GAP-003	Offset suggestion components	Offset integration has implementation evidence in the feature matrix, but no isolated component test row is currently documented.	Not covered	No CarbonOffset unit test appears in test/test-list.md.

Module Section: Community Impact Module

Objective: Validate community impact features from the feature matrix: eco-reviews, ratings, helpful votes, challenges, progress, and badges through frontend hook behavior.

Command: npm run test --workspace=frontend -- useCommunity

Result: 1 frontend hook file passed, 4 documented community tests passed, 0 failed.

Isolation and mocking: Community fetch calls, auth state, mutation responses, and QueryClient invalidation are controlled in hook-level tests.

Test Case ID	Unit Under Test	Scenario	Status	Evidence
UT-COMMUNITY-001	useCommunity	Load public community data.	Pass	Hook returned mocked community data.
UT-COMMUNITY-002	useCommunity	Create review and invalidate review/challenge data.	Pass	Mutation and invalidation calls matched.
UT-COMMUNITY-003	useCommunity	Toggle helpful votes.	Pass	Mutation request matched expected review action.
UT-COMMUNITY-004	useCommunity	Join challenges and update progress.	Pass	Challenge mutation behavior matched expected calls.

Module Section: Analytics and Reporting Module

Objective: Validate analytics and reporting features from the feature matrix: personal impact calculations, sustainability report source data, and goal-progress summary inputs where covered by isolated utilities.

Command: npm run test --workspace=backend -- src/utis/analytics.test.ts

Result: 1 file failed, 7 tests passed, 1 test failed.

Isolation and mocking: Analytics functions are called directly with controlled trip inputs and carbon-history rows. No database or HTTP route is used.

Test Case ID	Unit Under Test	Scenario	Status	Evidence
UT-ANALYTICS-	calculateTripImpact	Compare flight emissions against	Pass	Baseline and saved values matched expected close values.

Test Case ID	Unit Under Test	Scenario	Status	Evidence
001		economy-flight baseline.		
UT-ANALYTICS-002	calculateTripImpact	Compare rail emissions against petrol-car baseline.	Pass	Baseline and saved values matched expected close values.
UT-ANALYTICS-003	calculateTripImpact	Compare bus emissions against petrol-car baseline.	Pass	Baseline and saved values matched expected close values.
UT-ANALYTICS-004	calculateTripImpact	Compare hybrid car emissions against petrol-car baseline.	Pass	Baseline and saved values matched expected close values.
UT-ANALYTICS-005	calculateTripImpact	Compare taxi emissions against petrol-car baseline.	Pass	Baseline and saved values matched expected close values.
UT-ANALYTICS-006	calculateTripImpact	Compare budget hotel emissions against standard-hotel baseline.	Pass	Baseline and saved values matched expected close values.
UT-ANALYTICS-007	calculateTripImpact	Clamp saved emissions at zero when actual exceeds baseline.	Pass	savedKg was 0.
UT-ANALYTICS-008	buildImpactSummary	Summarize mixed entries with totals, breakdowns, details, and tree equivalent.	Fail	Expected carbonSavedKg 42 but received 41.9 at backend/src/utis/analytics.test.ts:57.
UT-ANALYTICS-GAP-001	PDF report generation UI	Sustainability report download behavior is implemented, but no isolated unit test row is currently documented for the PDF generation control path.	Not covered	No analytics page or PDF report component unit test appears in test/test-list.md.
UT-ANALYTICS-GAP-002	Goal setting UI/API helpers	Goal setting is implemented, but no isolated unit test row is currently documented for budget-goal setup or alert state behavior.	Not covered	No analytics goal component or hook unit test appears in test/test-list.md.

3. Command Output Evidence

Frontend unit run: `npm run test --workspace=frontend`

Result: Test Files 11 passed (11); Tests 52 passed (52); Duration 2.34s.

Backend unit run: `npm run test --workspace=backend -- src/utils/auth.test.ts src/utils/analytics.test.ts src/middleware/auth.test.ts`

Result: Test Files 1 failed, 2 passed (3); Tests 1 failed, 14 passed (15). Failure: backend/src/utils/analytics.test.ts expected summary.carbonSavedKg to be 42 but received 41.9.

4. Notes And Limitations

This report excludes backend integration suites: backend/src/routes/auth-profile.integration.test.ts and backend/src/routes/analytics.integration.test.ts. Those belong in the integration report because they exercise API and database boundaries.

Frontend tests use jsdom and mocked framework/data boundaries, so they validate component, hook, and utility behavior rather than a real browser or live API.

Backend authentication middleware tests use mocked Express objects, so they validate middleware branch behavior without starting the server.

The current failing analytics unit should be resolved before declaring the whole unit suite passed.